

Module 1: Reduced resistive MHD

The version-1 model is two-dimensional, incompressible, two-field reduced resistive magnetohydrodynamics on a doubly periodic Cartesian domain:

$$(x, y, t) \in [0, 2\pi) \times [0, 2\pi) \times [0, T].$$

$$\mathbf{q}(x, y, t) = (\psi, U), \quad \boldsymbol{\mu} = (\eta, \nu, A_0).$$

- ▶ ψ is the magnetic-flux function and U is the flow stream function.
- ▶ η is magnetic diffusivity, ν is kinematic viscosity, and A_0 sets the initial perturbation.
- ▶ The initial trajectories are synthetic reduced-MHD solutions, not genuine M3D-C1 output.

Two potentials define the physical fields

Magnetic field

$$\mathbf{B}_\perp = \nabla\psi \times \hat{\mathbf{z}} = (\psi_y, -\psi_x),$$

$$J = -\nabla^2\psi.$$

J is the out-of-plane current variable.

Fluid velocity

$$\mathbf{v}_\perp = \hat{\mathbf{z}} \times \nabla U = (-U_y, U_x),$$

$$\omega = \nabla^2 U.$$

ω is the out-of-plane vorticity.

$$\nabla \cdot \mathbf{B}_\perp = 0, \quad \nabla \cdot \mathbf{v}_\perp = 0.$$

Thus divergence constraints hold identically. The project predicts (ψ, U) and derives (J, ω) .

The Poisson bracket represents advection

$$[f, g] = f_x g_y - f_y g_x.$$

The stream-function velocity converts transport into a bracket:

$$\mathbf{v}_\perp \cdot \nabla f = (-U_y)f_x + (U_x)f_y = [U, f], \quad \frac{Df}{Dt} = f_t + [U, f].$$

$$[f, g] = -[g, f], \quad \int_\Omega [f, g] \, dA = 0,$$

Thus nonlinear brackets redistribute fields and exchange energy, but do not directly dissipate it.

Two PDEs govern the dynamics

$$\psi_t + [U, \psi] = \eta \nabla^2 \psi,$$

$$\omega_t + [U, \omega] = [J, \psi] + \nu \nabla^2 \omega.$$

- ▶ $[U, \psi]$ transports magnetic-flux contours with the fluid.
- ▶ $\eta \nabla^2 \psi = -\eta J$ permits resistive slippage and topology change in current layers.
- ▶ $[U, \omega]$ transports vorticity, while $[J, \psi]$ is magnetic forcing from the Lorentz force.
- ▶ $\nu \nabla^2 \omega$ smooths vorticity and dissipates kinetic energy.

Alfvén normalization sets the scales

$$V_A = \frac{B_0}{\sqrt{\mu_0 \rho_0}}, \quad t_A = \frac{L_0}{V_A}.$$

$$\eta = \frac{\eta_m}{L_0 V_A}, \quad \nu = \frac{\nu_d}{L_0 V_A}.$$

$$S = \frac{1}{\eta} = \frac{t_\eta}{t_A}, \quad \text{Re} = \frac{1}{\nu} = \frac{t_\nu}{t_A}, \quad \text{Pm} = \frac{\nu}{\eta}.$$

S and Re compare resistive and viscous diffusion times with t_A . Neural standardization is a later numerical transformation, not physical nondimensionalization.

Periodicity and energy close the model

Periodic faces and PDE-required derivatives match. Gauge and mean-flux conditions are

$$\boxed{\langle U \rangle(t) = 0}, \quad \frac{d}{dt} \langle \psi \rangle = 0.$$

The exact periodic energy law is

$$\boxed{\frac{d}{dt} \frac{1}{2} \int_{\Omega} (|\nabla \psi|^2 + |\nabla U|^2) dA = -\eta \int_{\Omega} J^2 dA - \nu \int_{\Omega} \omega^2 dA \leq 0.}$$

Finite η enables reconnection in current sheets. This is not full M3D-C1 equivalence.

Module 4: A data-only field surrogate

The parameterized coordinate network approximates a family of fields:

$$\mathcal{N}_{\theta} : (x, y, t, \eta, \nu, A_0) \mapsto (\hat{\psi}, \hat{U}).$$

Data are split by complete parameter cases:

$$\mu_c = (\eta_c, \nu_c, A_{0,c}).$$

- ▶ No point from a held-out case enters training or preprocessing.
- ▶ Parameter interpolation and extrapolation are separate; interpolation within $t \in [0, T]$ is not forecasting beyond T .

Periodic inputs respect the domain

$$\phi_x = \frac{2\pi x}{L_x}, \quad \phi_y = \frac{2\pi y}{L_y},$$

$$\mathbf{z} = (\sin \phi_x, \cos \phi_x, \sin \phi_y, \cos \phi_y, \tilde{t}, \tilde{r}_\eta, \tilde{r}_\nu, \tilde{A}_0).$$

Smooth functions of $\mathbf{z}$ are periodic in value and derivatives. Scaling statistics come from training cases only.

Baseline: six fully connected tanh layers of width 128 and a linear two-output head. Module 5 retains the same architecture family and periodic representation.

Training uses labeled field error only

For standardized output errors

$$e_{\psi,n} = \hat{\tilde{\psi}}_n - \tilde{\psi}_n, \quad e_{U,n} = \hat{\tilde{U}}_n - \tilde{U}_n,$$

the case-balanced objective is

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{data}} = \frac{1}{|\mathcal{C}_{\mathcal{B}}|} \sum_{c \in \mathcal{C}_{\mathcal{B}}} \left[\frac{1}{N_c} \sum_{n \in c} e_{\psi,n}^2 + \frac{1}{N_c} \sum_{n \in c} e_{U,n}^2 \right].$$

There is no PDE, initial-condition, boundary-condition, gauge, global-balance, J , ω , derivative, or spectral target loss. Adam, mini-batches, regularization, and early stopping are optimization choices; they do not make the model physics-informed.

The baseline makes the PINN testable

Evaluation uses complete unseen fields, not training loss alone. For $a \in \{\psi, U, J, \omega\}$,

$$\varepsilon_{L_2}^{(a)} = \frac{\|\hat{a} - a\|_2}{\|a\|_2 + \epsilon}.$$

$$\hat{J} = -\nabla^2 \hat{\psi}, \quad \hat{\omega} = \nabla^2 \hat{U},$$

Derived fields, energy, reconnection measures, periodicity, and PDE residuals are evaluation-only diagnostics.

Controlled comparison: Modules 4 and 5 share splits, labels, preprocessing, periodic encoding, architecture family, data loss, metrics, and seeds. Physics helps only if the matched PINN improves a declared metric, with added compute disclosed.

Module 5: PDE residuals enter training

The network still predicts $(\hat{\psi}, \hat{U})$, while automatic differentiation constructs

$$\hat{J} = -\nabla^2 \hat{\psi}, \quad \hat{\omega} = \nabla^2 \hat{U}.$$

The local reduced-MHD equation defects are

$$R_{\psi} = \hat{\psi}_t + [\hat{U}, \hat{\psi}] - \eta \nabla^2 \hat{\psi},$$

$$R_{\omega} = \hat{\omega}_t + [\hat{U}, \hat{\omega}] - [\hat{J}, \hat{\psi}] - \nu \nabla^2 \hat{\omega}.$$

At unlabeled collocation points, the PINN receives no solver field values; it is penalized when these residuals are nonzero.

The objective combines five constraints

$$\mathcal{L}_{\text{total}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{PDE}}\mathcal{L}_{\text{PDE}} + w_{\text{IC}}\mathcal{L}_{\text{IC}} + w_{\text{BC}}\mathcal{L}_{\text{BC}} + w_{\text{global}}\mathcal{L}_{\text{global}}.$$

- ▶ $\mathcal{L}_{\text{data}}$: the identical labeled observations used by Module 4.
- ▶ $\mathcal{L}_{\text{PDE}}$: normalized mean-square R_{ψ} and R_{ω} at collocation points.
- ▶ $\mathcal{L}_{\text{IC}}$: selects the intended trajectory at $t = 0$.
- ▶ $\mathcal{L}_{\text{BC}} = 0$ for hard periodicity; $\mathcal{L}_{\text{global}}$ contains gauge, mean-flux, and optional balances.

The vorticity residual reaches fourth order

Because $\hat{\omega} = \nabla^2 \hat{U}$,

$$\hat{\omega}_t \sim \partial_t \partial_x^2 \hat{U}, \quad [\hat{U}, \hat{\omega}] \sim \partial_x^3 \hat{U}, \quad \nabla^2 \hat{\omega} = \nabla^4 \hat{U}.$$

The Lorentz bracket also contains third derivatives of $\hat{\psi}$. Therefore smooth activations such as tanh are required, scaling factors must remain inside the differentiation graph, and derivative precision must be tested.

The PDE cannot identify a spatially constant offset in U , so a gauge condition remains necessary:

$$\langle \hat{U} \rangle(t, \mu) = 0.$$

Balances and curricula stabilize training

$$\frac{d}{dt} \frac{1}{2} \int_{\Omega} (|\nabla \hat{\psi}|^2 + |\nabla \hat{U}|^2) dA + \eta \int_{\Omega} \hat{J}^2 dA + \nu \int_{\Omega} \hat{\omega}^2 dA = 0.$$

Collocation defines the residual norm minimized. Uniform coverage may be supplemented by time-aware, structure-aware, or residual-adaptive points; validation points remain fixed.

Training proceeds in stages:

derivative tests $\rightarrow$ data warm start $\rightarrow R_{\psi} \rightarrow R_{\omega}$
 $\rightarrow$ time curriculum $\rightarrow$ parameterized cases.

Global penalties follow quadrature verification.

Physics must add measurable value

Module 6 compares matched models on unseen cases using field errors, residuals, condition defects, global balances, and cost.

- ▶ A PINN may improve labeled-data efficiency, derivative fidelity, parameter generalization, or physical admissibility.
- ▶ It may also train more slowly, oversmooth current sheets, suffer gradient conflict, or inherit error from an incomplete physical model.
- ▶ Reports include repeated seeds, failures, extrapolation, and data-versus-compute fairness.

Scientific conclusion: the PINN is not assumed to win. The result remains a surrogate of synthetic two-field reduced-MHD dynamics; an M3D-C1 surrogate requires genuine M3D-C1 data and Module 7.